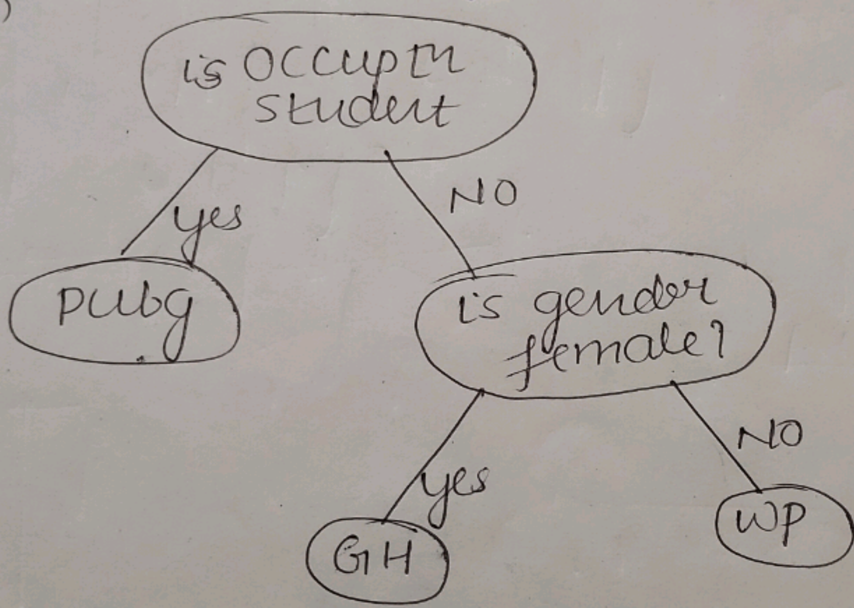
DECISION TREE 01

→ nested if else condition

eg:) 01

Gender	occupⁿ	suggestion
F	student	PUBG
F	programm	Github
M	Pg	WP
F	Pg	GH
M	student	Pubg
M	student	Pubg

tree:)

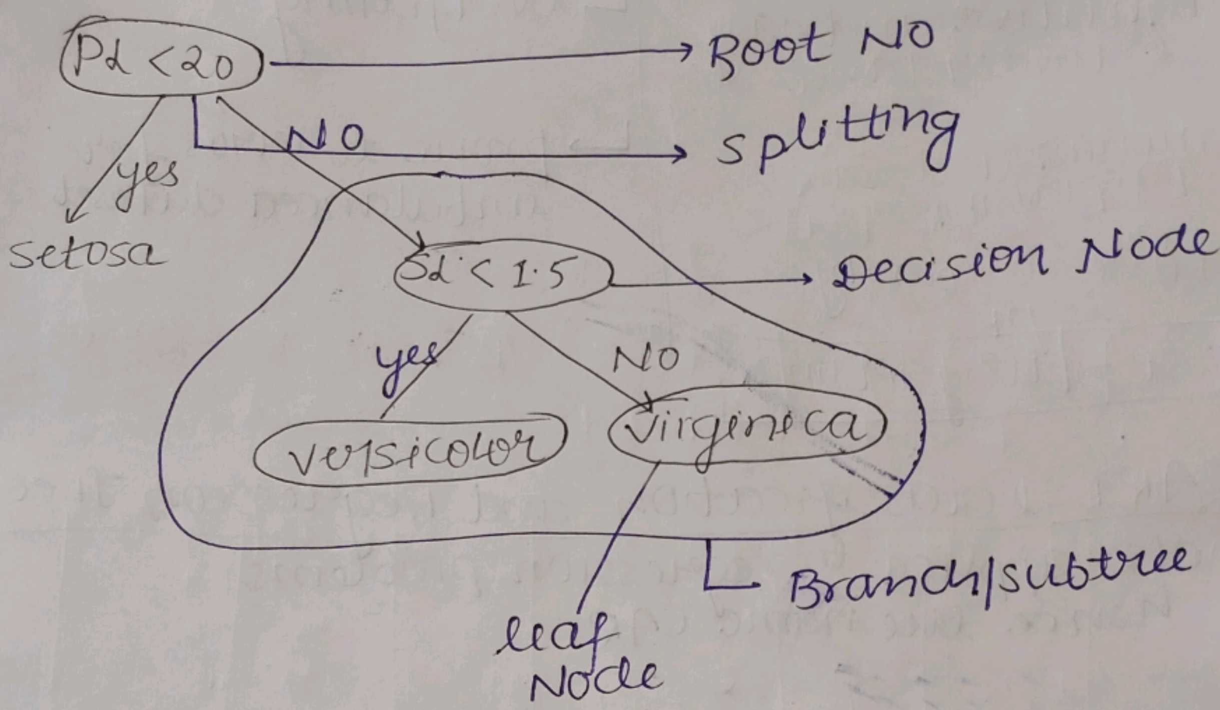


Conclusion

a) Decision trees are nothing but nested if-else giant condition

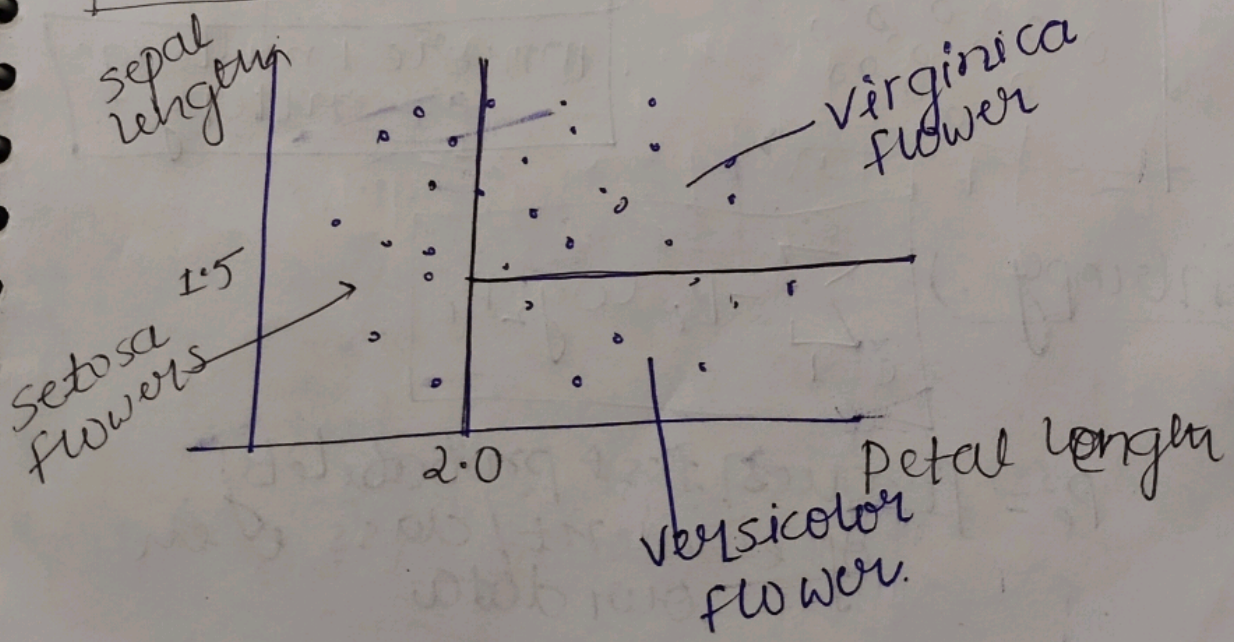
b) Decision trees use hyperplanes which run parllel to any one of axes to cut your cordinate system into hyper cuboids

Termonology



Q.}

Petal length	sepal length	Type
1.34	0.34	setosa
3.45	1.45	versicolor
1.69	0.98	setosa
2.56	1.79	virginica
3.0	1.13	versicolor
1.3	0.88	setosa

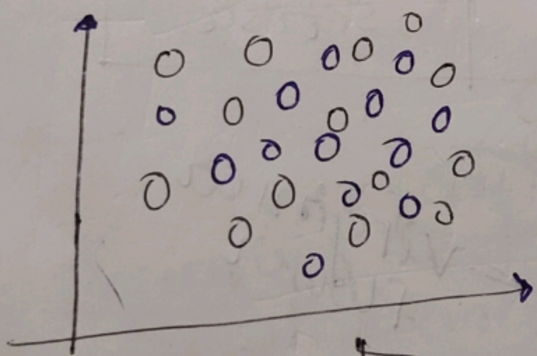


Advantage	Disadvantage
Intuitive and easy to understand	↳ overfitting
minimal data prepration	↳ prone to errors for imbalanced dataset
Time complexity = logn	
$\boxed{\text{cost} \Rightarrow \text{alogrithmic}}$	

CART :) Classification and Regression Tree
also applied to regression problems
hence the name CART

⊕ Decision tree Entropy

what is Entropy?
measure of disorder or measure of impurity or purity



$\boxed{\text{⊕ more knowledge less enttopy}}$

Entropy :) $\boxed{\sum_{i=1}^{c} -P_i \log_2 P_i}$

P_i = frequentist probability of element/class i in our data

$$\boxed{E(D) = -P_{yes}\log_2(P_{yes}) - P_{no}\log_2(P_{no})}$$

Q calculate the entropy

salary	age	purchase
2000	21	yes
10,000	45	NO
60,000	27	yes
1500	31	NO
1200	18	NO

sal	age	purch
3400	31	NO
15000	25	NO
69000	57	yes
25000	21	NO
32000	28	NO

Q) entropy

for Data-I

$P_{yes} = P_{yes}\log(P_y)$

$P_y = \frac{2}{5}$

$P_{yes} = \frac{2}{5}\log(\frac{2}{5})$

$P_n = \frac{3}{5}\log\frac{3}{5}$

entropy $= -P_y\log y - P_n\log n$

$\Rightarrow \boxed{-\frac{2}{5}\log(2/5) - \frac{3}{5}\log(3/5)}$

entropy = ~~[illegible]~~

$H(d) = 0.97$

for Data-2

$P_y = P_y\log y$

$= \frac{1}{5}\log 1/5$

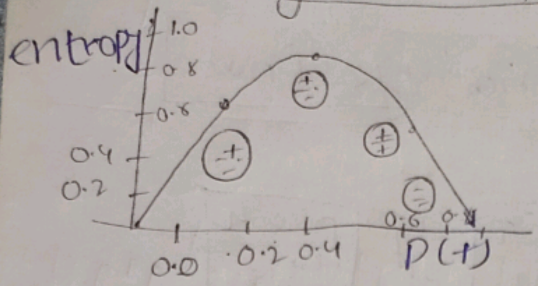
$P_n = \frac{4}{5}\log 4/5$

$$\boxed{P_E = -\frac{1}{5}\log\frac{1}{5} - \frac{4}{5}\log(\frac{4}{5})}$$

$H(d) = 0.72$

→ more entropy more uncertainity
↳ for a 2 class ~~problem~~ problem the min entropy is 0 and the max is 1
  ↳ not for more than 2 class
↳ you can use $\log_2$ or $\log_e$ can be use to calculate.

Entropy vs probability



Entropy for continous variables

method :) 01 short cut

Dataset-1

Area	Built	price
1200	1999	35
1800	2011	56
1400	2000	7.3

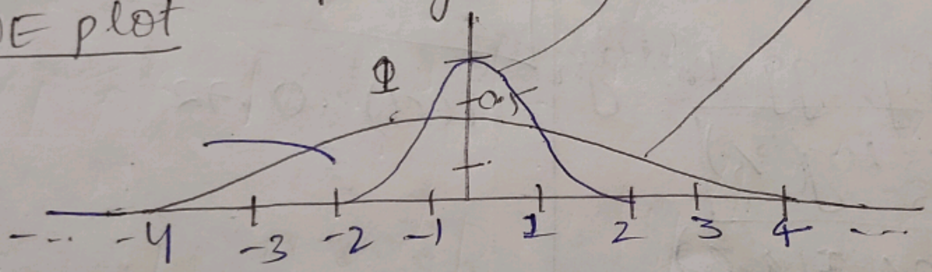
plot graph

Dataset2

Area	Built	price
2200	1989	4.6
800	2018	6.5
1100	2005	12.8

plot graph

KDE plot



for Dataset-2 has more knowledge as more flattern in x

so, $\boxed{\text{knowledge} \propto \frac{1}{\text{entropy}}}$

Information Gain

- matrix used to train Decission tree
  - ↳ this metric measures the quality of a split.
  - ↳ Information gain is based on decrease in entropy after a data-set is split in an attribute
  - ↳ Construction a decision tree is all about finding attribute that return the highest information gain

$$\boxed{IG = E(\text{parent}) - \{\text{weight Avg}\} * E\{\text{children}\}}$$

- steps for calculating information Gain

Step 1

↳ Entropy of parent

$$E(P) = -P_y \log_2(P_y) - P_n \log_2(P_n)$$

$$E(P) = -\frac{9}{14}\log_2(9/14) - \frac{5}{14}\log_2(5/14)$$

yes = 9
N = 5
total = 14

$$\boxed{E(P) = 0.94}$$

Step 2 Calculate of Entropy for children

$$E(\text{child } 1) = -\frac{2}{5}\log\left(\frac{2}{5}\right)$$

child 1

-1

child 2

child 3.

Step 3. calculate weight entropy of children

as one's entropy 0 then its leaf Node

Step 4 calculate information gain

$$I \cdot G = E(\text{parent}) - \{\text{weighted avg}\} * E(\text{children})$$

Step 5 ~~I.G~~
calculate information gain for all the column

→ which column has highest information gain the algorithm will select that column to split the data.

Step :) 6 find information Gain recursively

Gini impurity

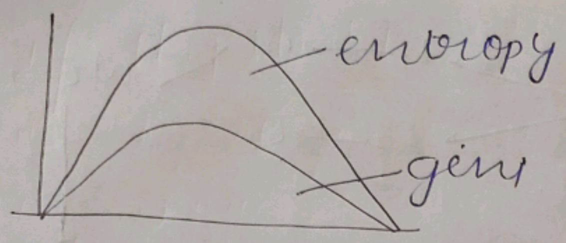
A way to measure the impurity

$$\boxed{G_I = 1 - P_y^2 + P_x^2}$$

⊕ behavious is same as entropy

Difference

(4)



Handling numerical data

Step 1) sort the data set on the numerical basics of numerical column.

Step 2) split the entire data on the basics of every value of user-rating

Step 3

$D \rightarrow (f > v_1) \rightarrow D_1, D_2$

$D \rightarrow (\text{rating} > 1.9) \rightarrow$ Dataset 1, Dataset 2

simillary

$D \rightarrow (f > v_2) \rightarrow D_1, D_2$

$D \rightarrow (f > v_3) \rightarrow D_1, D_2$

D = Datset
f = colum of user-rating
$v_1, v_2 .. v_n$ are the various rows of user Rating

Step (4)) calculate entropy for all child.

$D \rightarrow (f > v_1) \rightarrow D_1 \rightarrow E_1,\ D_2 \rightarrow E_2 \rightarrow$ weigted entropy $\rightarrow IG_1$

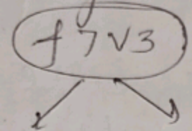
$D \rightarrow (f > v_2)$ → $D_1 \rightarrow G$; $D_2 \rightarrow G_2$ > WEG → IG2

WEG ↓ Calculate the information gain of all

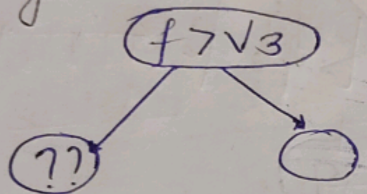
Step 5

max{IG1, IG2, IG3 ... IGn}

we got this



Step 6 Do this recursively until you get all the above node.



Q: Entire things look computationally too expensive, is the right way of finding the splitting criteria?

→ yes

train time complexity is high

test time complexity is low

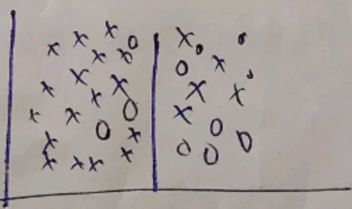
Hyperparameters
└→ means tuning nobs

I ⓐ overfitting
performwell → training
perform badly → Test

A) max_depth = None
can be ~~reach~~ reach to outliear.
and cause overfit

└→ outliear

II ⓑ underfitting
└→ don't understand the complexity



B Criteria:) ⟶ entropy or gini] → both like are same.

c) splitter:) for numberic
└ to reduce overfitting

d) ~~???~~
d) min sample split
ⓐ lower value
└ overfitting
ⓑ higher value
└ underfitting

e) min sample leaf

f) max feature :) How much column you want it.
↳ reduce number of column

g) max leaf nodes
↳ control How much leaf node you want.

⊙ High value → overfit
lower value → underfit

h) min impurity decrease :)

Variants/algorithm you should known

1. ID3
2. C4.5
3. CART
4. CHAID

1. ID3 :) earliest, D.T, choose attribute give highest Information Gain.

2. C4.5 :) Improved one, It uses Gain Ratio
↳ Handle missing value
↳ perform pruning

3. CART (Classification and Regression Tree)
Binary split

d). CHAID

Chai-square Automatic Interaction Detection

↳ multiple way split